

AFAR REGIONAL PORTAL V2

Database Architecture & MySQL 8 Schema Specification — Prepared by Senior Database Engineer

1. Executive Summary & Design Core Decisions

This specification delivers a fully modernized, high-integrity MySQL 8 database schema layout for the **Afar Regional Portal v2**. It addresses architectural vulnerabilities discovered during previous audits—including missing foreign key integrity, unindexed filter targets, and highly fragmented internationalization (i18n) schemas—while strictly preserving the public navigation structures required by regional citizens.

Key Structural Principles

- **Unified Polymorphic Media Engine:** Replaces isolated mechanisms (such as `NewsImage` and incomplete `MultiImages`) with a single polymorphic media structure mapped using `mediable_type` and `mediable_id`.
- **Native Bilingual JSON Storage:** Consolidates duplicate column pairs (e.g., `title` and `titleamh`) into unified, single JSON columns cast as native text maps (e.g., `{"en": "...", "am": "..."}`). This design simplifies multi-language indexing and allows future language variations to be added without necessitating DDL structural schema updates.
- **Strict Referential Integrity:** Enforces standard MySQL InnoDB relational engine constraints via explicitly defined data types, utilizing `CASCADE` or `RESTRICT` clauses precisely depending on entity dependency trees.
- **Comprehensive Soft Deletes:** Implements uniform `deleted_at` timestamp tracking columns across all operational tables to safeguard historical and auditable governmental records.

Optimization Strategy: To circumvent MySQL restrictions prohibiting direct `FULLTEXT` indexes on JSON structures, this schema implements **Generated Virtual Columns** to automatically extract multi-language content, applying optimized indexing structures to these virtual layers.

2. Complete Migration Order & Execution Plan

Tables must be initialized sequentially to honor foreign key dependency restrictions. The setup order proceeds from structural independent catalogs to dependent transactional modules:

Sequence	Table(s) Name	Dependency Profile	Purpose / Architectural Scope
1	users	Independent Baseline	Staff account core identity storage.
2	roles, permissions	Independent Baseline	RBAC baseline structures powered by

Sequence	Table(s) Name	Dependency Profile	Purpose / Architectural Scope
			Spatie conventions.
3	model_has_roles, model_has_permissions, role_has_permissions	Depends on Users/RBAC	Spatie RBAC relationship join mappings.
4	categories	Independent Lookup	Unified lookups for Directorates and City Administrations.
5	about, sliders, galleries, announcements, contacts, subscribers	Independent Modules	Static pages, structural configurations, public inputs, newsletters.
6	news, events, tenders, vacancies, publications	Depends on Users	Core portal transactional modules containing author audit ties.
7	directorates, city_admins	Depends on Categories	Regional administrative data linked directly to Categories.
8	media	Polymorphic Layer	Unified cross-module attachment management matrix.
9	activity_log	Global Auditor	System-wide activity tracking linking to all systemic actors.

3. MySQL DDL Schema Definition

The following raw SQL defines the target database schema structure, incorporating optimized indexes and storage attributes:

```
-- =====
-- PHASE 1: CORE RBAC AND LOOKUPS
-- =====

CREATE TABLE `users` (
  `id` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  `name` VARCHAR(255) NOT NULL,
  `email` VARCHAR(255) NOT NULL UNIQUE,
  `password` VARCHAR(255) NOT NULL,
  `created_at` TIMESTAMP NULL DEFAULT NULL,
  `updated_at` TIMESTAMP NULL DEFAULT NULL,
  `deleted_at` TIMESTAMP NULL DEFAULT NULL,
  INDEX `idx_users_deleted` (`deleted_at`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

CREATE TABLE `roles` (
  `id` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  `name` VARCHAR(255) NOT NULL,
  `guard_name` VARCHAR(255) NOT NULL,
  `created_at` TIMESTAMP NULL DEFAULT NULL,
  `updated_at` TIMESTAMP NULL DEFAULT NULL,
  UNIQUE KEY `unique_role_guard` (`name`, `guard_name`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

CREATE TABLE `permissions` (
  `id` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  `name` VARCHAR(255) NOT NULL,
  `guard_name` VARCHAR(255) NOT NULL,
  `created_at` TIMESTAMP NULL DEFAULT NULL,
  `updated_at` TIMESTAMP NULL DEFAULT NULL,
  UNIQUE KEY `unique_permission_guard` (`name`, `guard_name`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

CREATE TABLE `model_has_roles` (
  `role_id` BIGINT UNSIGNED NOT NULL,
  `model_type` VARCHAR(255) NOT NULL,
  `model_id` BIGINT UNSIGNED NOT NULL,
  PRIMARY KEY (`role_id`, `model_id`, `model_type`),
  INDEX `idx_model_role` (`model_id`, `model_type`),
  CONSTRAINT `fk_model_roles_role` FOREIGN KEY (`role_id`) REFERENCES `roles` (`id`) ON
  DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

CREATE TABLE `model_has_permissions` (
  `permission_id` BIGINT UNSIGNED NOT NULL,
  `model_type` VARCHAR(255) NOT NULL,
  `model_id` BIGINT UNSIGNED NOT NULL,
  PRIMARY KEY (`permission_id`, `model_id`, `model_type`),
```

```

    INDEX `idx_model_permission` (`model_id`, `model_type`),
    CONSTRAINT `fk_model_perms_perm` FOREIGN KEY (`permission_id`) REFERENCES `permissions`
(`id`) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

CREATE TABLE `role_has_permissions` (
    `permission_id` BIGINT UNSIGNED NOT NULL,
    `role_id` BIGINT UNSIGNED NOT NULL,
    PRIMARY KEY (`permission_id`, `role_id`),
    CONSTRAINT `fk_role_perms_perm` FOREIGN KEY (`permission_id`) REFERENCES `permissions`
(`id`) ON DELETE CASCADE,
    CONSTRAINT `fk_role_perms_role` FOREIGN KEY (`role_id`) REFERENCES `roles` (`id`) ON
DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

CREATE TABLE `categories` (
    `id` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    `type` ENUM('directoriate', 'city_admin') NOT NULL,
    `name` JSON NOT NULL,
    `slug` VARCHAR(255) NOT NULL UNIQUE,
    `created_at` TIMESTAMP NULL DEFAULT NULL,
    `updated_at` TIMESTAMP NULL DEFAULT NULL,
    `deleted_at` TIMESTAMP NULL DEFAULT NULL,
    INDEX `idx_categories_lookup` (`type`, `deleted_at`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

-- =====
-- PHASE 2: CORE PORTAL CHANNELS & ENTITIES
-- =====

CREATE TABLE `news` (
    `id` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    `user_id` BIGINT UNSIGNED NOT NULL,
    `title` JSON NOT NULL,
    `body` JSON NOT NULL,
    `status` ENUM('draft', 'published') NOT NULL DEFAULT 'draft',
    `published_at` TIMESTAMP NULL DEFAULT NULL,
    `created_at` TIMESTAMP NULL DEFAULT NULL,
    `updated_at` TIMESTAMP NULL DEFAULT NULL,
    `deleted_at` TIMESTAMP NULL DEFAULT NULL,

    -- Virtual Extracted Columns for Search Indexing
    `title_en` VARCHAR(255) GENERATED ALWAYS AS (JSON_UNQUOTE(JSON_EXTRACT(`title`,
'$$.en')))) STORED,
    `title_am` VARCHAR(255) GENERATED ALWAYS AS (JSON_UNQUOTE(JSON_EXTRACT(`title`,
'$$.am')))) STORED,

    CONSTRAINT `fk_news_author` FOREIGN KEY (`user_id`) REFERENCES `users` (`id`) ON DELETE
RESTRICT,
    INDEX `idx_news_query` (`status`, `published_at`, `deleted_at`),
    FULLTEXT INDEX `ft_news_bilingual` (`title_en`, `title_am`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

```

```

CREATE TABLE `events` (
  `id` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  `title` JSON NOT NULL,
  `start_at` TIMESTAMP NOT NULL,
  `end_at` TIMESTAMP NOT NULL,
  `created_at` TIMESTAMP NULL DEFAULT NULL,
  `updated_at` TIMESTAMP NULL DEFAULT NULL,
  `deleted_at` TIMESTAMP NULL DEFAULT NULL,
  INDEX `idx_events_range` (`start_at`, `end_at`, `deleted_at`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

CREATE TABLE `tenders` (
  `id` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  `title` JSON NOT NULL,
  `body` JSON NOT NULL,
  `file_path` VARCHAR(255) NOT NULL,
  `opens_at` TIMESTAMP NOT NULL,
  `closes_at` TIMESTAMP NOT NULL,
  `created_at` TIMESTAMP NULL DEFAULT NULL,
  `updated_at` TIMESTAMP NULL DEFAULT NULL,
  `deleted_at` TIMESTAMP NULL DEFAULT NULL,

  `title_en` VARCHAR(255) GENERATED ALWAYS AS (JSON_UNQUOTE(JSON_EXTRACT(`title`,
'$$.en')))) STORED,
  `title_am` VARCHAR(255) GENERATED ALWAYS AS (JSON_UNQUOTE(JSON_EXTRACT(`title`,
'$$.am')))) STORED,

  INDEX `idx_tenders_timeline` (`opens_at`, `closes_at`, `deleted_at`),
  FULLTEXT INDEX `ft_tenders_title` (`title_en`, `title_am`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

CREATE TABLE `vacancies` (
  `id` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  `title` JSON NOT NULL,
  `body` JSON NOT NULL,
  `position` JSON NOT NULL,
  `salary` VARCHAR(255) NULL DEFAULT NULL,
  `file_path` VARCHAR(255) NOT NULL,
  `created_at` TIMESTAMP NULL DEFAULT NULL,
  `updated_at` TIMESTAMP NULL DEFAULT NULL,
  `deleted_at` TIMESTAMP NULL DEFAULT NULL,

  `title_en` VARCHAR(255) GENERATED ALWAYS AS (JSON_UNQUOTE(JSON_EXTRACT(`title`,
'$$.en')))) STORED,
  `title_am` VARCHAR(255) GENERATED ALWAYS AS (JSON_UNQUOTE(JSON_EXTRACT(`title`,
'$$.am')))) STORED,

  INDEX `idx_vacancies_active` (`deleted_at`),
  FULLTEXT INDEX `ft_vacancies_title` (`title_en`, `title_am`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

CREATE TABLE `publications` (
  `id` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,

```

```

`title` JSON NOT NULL,
`file_path` VARCHAR(255) NOT NULL,
`created_at` TIMESTAMP NULL DEFAULT NULL,
`updated_at` TIMESTAMP NULL DEFAULT NULL,
`deleted_at` TIMESTAMP NULL DEFAULT NULL,

`title_en` VARCHAR(255) GENERATED ALWAYS AS (JSON_UNQUOTE(JSON_EXTRACT(`title`,
'$$.en')))) STORED,
`title_am` VARCHAR(255) GENERATED ALWAYS AS (JSON_UNQUOTE(JSON_EXTRACT(`title`,
'$$.am')))) STORED,

INDEX `idx_publications_active` (`deleted_at`),
FULLTEXT INDEX `ft_publications_title` (`title_en`, `title_am`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

CREATE TABLE `announcements` (
  `id` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  `title` JSON NOT NULL,
  `body` JSON NOT NULL,
  `created_at` TIMESTAMP NULL DEFAULT NULL,
  `updated_at` TIMESTAMP NULL DEFAULT NULL,
  `deleted_at` TIMESTAMP NULL DEFAULT NULL,

  `title_en` VARCHAR(255) GENERATED ALWAYS AS (JSON_UNQUOTE(JSON_EXTRACT(`title`,
'$$.en')))) STORED,
  `title_am` VARCHAR(255) GENERATED ALWAYS AS (JSON_UNQUOTE(JSON_EXTRACT(`title`,
'$$.am')))) STORED,

  INDEX `idx_announcements_active` (`deleted_at`),
  FULLTEXT INDEX `ft_announcements_title` (`title_en`, `title_am`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

-- =====
-- PHASE 3: ADMINISTRATIVE STRUCTURES & CONFIGURATIONS
-- =====

CREATE TABLE `about` (
  `id` INT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  `title` JSON NOT NULL,
  `body` JSON NOT NULL,
  `updated_at` TIMESTAMP NULL DEFAULT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

CREATE TABLE `sliders` (
  `id` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  `title` JSON NULL DEFAULT NULL,
  `image_path` VARCHAR(255) NOT NULL,
  `sort_order` INT NOT NULL DEFAULT 0,
  `created_at` TIMESTAMP NULL DEFAULT NULL,
  `updated_at` TIMESTAMP NULL DEFAULT NULL,
  `deleted_at` TIMESTAMP NULL DEFAULT NULL,
  INDEX `idx_sliders_sort` (`sort_order`, `deleted_at`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

```

```

CREATE TABLE `galleries` (
  `id` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  `title` JSON NOT NULL,
  `path` VARCHAR(255) NOT NULL,
  `type` ENUM('photo', 'video') NOT NULL,
  `created_at` TIMESTAMP NULL DEFAULT NULL,
  `updated_at` TIMESTAMP NULL DEFAULT NULL,
  `deleted_at` TIMESTAMP NULL DEFAULT NULL,
  INDEX `idx_galleries_filter` (`type`, `deleted_at`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

CREATE TABLE `directorates` (
  `id` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  `category_id` BIGINT UNSIGNED NOT NULL,
  `title` JSON NOT NULL,
  `body` JSON NOT NULL,
  `file_path` VARCHAR(255) NULL DEFAULT NULL,
  `created_at` TIMESTAMP NULL DEFAULT NULL,
  `updated_at` TIMESTAMP NULL DEFAULT NULL,
  `deleted_at` TIMESTAMP NULL DEFAULT NULL,
  CONSTRAINT `fk_directorates_category` FOREIGN KEY (`category_id`) REFERENCES
`categories` (`id`) ON DELETE RESTRICT,
  INDEX `idx_directorates_lookup` (`category_id`, `deleted_at`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

CREATE TABLE `city_admins` (
  `id` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  `category_id` BIGINT UNSIGNED NOT NULL,
  `title` JSON NOT NULL,
  `body` JSON NOT NULL,
  `file_path` VARCHAR(255) NULL DEFAULT NULL,
  `created_at` TIMESTAMP NULL DEFAULT NULL,
  `updated_at` TIMESTAMP NULL DEFAULT NULL,
  `deleted_at` TIMESTAMP NULL DEFAULT NULL,
  CONSTRAINT `fk_city_admins_category` FOREIGN KEY (`category_id`) REFERENCES
`categories` (`id`) ON DELETE RESTRICT,
  INDEX `idx_city_admins_lookup` (`category_id`, `deleted_at`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

-- =====
-- PHASE 4: ENGAGEMENT, SYSTEM AUDITING & POLYMORPHIC MEDIA
-- =====

CREATE TABLE `contacts` (
  `id` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  `name` VARCHAR(255) NOT NULL,
  `email` VARCHAR(255) NOT NULL,
  `message` TEXT NOT NULL,
  `status` ENUM('new', 'read', 'replied') NOT NULL DEFAULT 'new',
  `created_at` TIMESTAMP NULL DEFAULT NULL,
  `updated_at` TIMESTAMP NULL DEFAULT NULL,
  `deleted_at` TIMESTAMP NULL DEFAULT NULL,

```

```

INDEX `idx_contacts_status` (`status`, `deleted_at`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

CREATE TABLE `subscribers` (
  `id` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  `email` VARCHAR(255) NOT NULL UNIQUE,
  `category` ENUM('all', 'tenders', 'vacancies') NOT NULL DEFAULT 'all',
  `verified_at` TIMESTAMP NULL DEFAULT NULL,
  `created_at` TIMESTAMP NULL DEFAULT NULL,
  `updated_at` TIMESTAMP NULL DEFAULT NULL,
  `deleted_at` TIMESTAMP NULL DEFAULT NULL,
  INDEX `idx_subscribers_alert` (`category`, `verified_at`, `deleted_at`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

CREATE TABLE `media` (
  `id` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  `mediable_type` VARCHAR(255) NOT NULL,
  `mediable_id` BIGINT UNSIGNED NOT NULL,
  `path` VARCHAR(255) NOT NULL,
  `type` ENUM('image', 'document') NOT NULL DEFAULT 'image',
  `sort_order` INT NOT NULL DEFAULT 0,
  `created_at` TIMESTAMP NULL DEFAULT NULL,
  `updated_at` TIMESTAMP NULL DEFAULT NULL,
  `deleted_at` TIMESTAMP NULL DEFAULT NULL,
  INDEX `idx_polymorphic_media` (`mediable_type`, `mediable_id`, `sort_order`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

CREATE TABLE `activity_log` (
  `id` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  `log_name` VARCHAR(255) NULL DEFAULT NULL,
  `description` TEXT NOT NULL,
  `subject_type` VARCHAR(255) NULL DEFAULT NULL,
  `subject_id` BIGINT UNSIGNED NULL DEFAULT NULL,
  `causer_type` VARCHAR(255) NULL DEFAULT NULL,
  `causer_id` BIGINT UNSIGNED NULL DEFAULT NULL,
  `properties` JSON NULL DEFAULT NULL,
  `batch_uuid` CHAR(36) NULL DEFAULT NULL,
  `created_at` TIMESTAMP NULL DEFAULT NULL,
  `updated_at` TIMESTAMP NULL DEFAULT NULL,
  INDEX `idx_activity_subject` (`subject_type`, `subject_id`),
  INDEX `idx_activity_causer` (`causer_type`, `causer_id`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

```

4. Entity Relationships & Referential Constraints

The integrity model utilizes standard parent-child constraints to maintain transactional health across workflows:

- **categories** → **directorates / city_admins**: Enforced via `ON DELETE RESTRICT`. This ensures categories linked to structural administrative records cannot be accidentally deleted from administration workflows.
- **users** → **news**: Enforced via `ON DELETE RESTRICT`. Retains strict audit tracking loops. User accounts linked to existing articles cannot be dropped; items must be re-assigned or the user soft-deleted instead.

- **Polymorphic Mappings (media & activity_log):** Regulated directly by application business logic layer observers to prevent hard lock conflicts across mixed entity types.
- **Spatie RBAC Tables:** Configured using `ON DELETE CASCADE` rules to automatically prune relationship links when core role or permission identities are modified.

5. Database Implementation & Optimization Plan

Index Topography

Standard sequential tracking keys are supplemented with specialized structural optimization components to guarantee sub-millisecond retrieval profiles:

- **Composite Query Layouts:** Configured on frequently filtered target fields, such as `idx_news_query` (`status`, `published_at`, `deleted_at`), preventing full table scans when handling public front-facing streams.
- **Generated Stored Columns + Full-Text Search:** By transforming complex internal JSON elements into static, flat fields (`title_en`, `title_am`), the application bypasses expensive real-time computational overhead and allows high-performance `FULLTEXT MATCH AGAINST` text search routines.

Data Migration Plan from Legacy Schema

1. **Parallel Prototyping environment setup:** Initialize the target layout side-by-side with the old production structure.
2. **Transformational Mapping Execution:** Execute script procedures to unpack duplicate legacy language blocks into structural arrays:

```
INSERT INTO news (id, user_id, title, body, status, published_at)
SELECT id, 1, JSON_OBJECT('en', title, 'am', titleamh), JSON_OBJECT('en', body, 'am', bodyamh), 'published', created_at
FROM legacy_database.news;
```

3. **Media Re-indexing Matrix:** Gather values from legacy multi-image records and inject them into the polymorphic media grid layout with explicit sort indexing parameters.